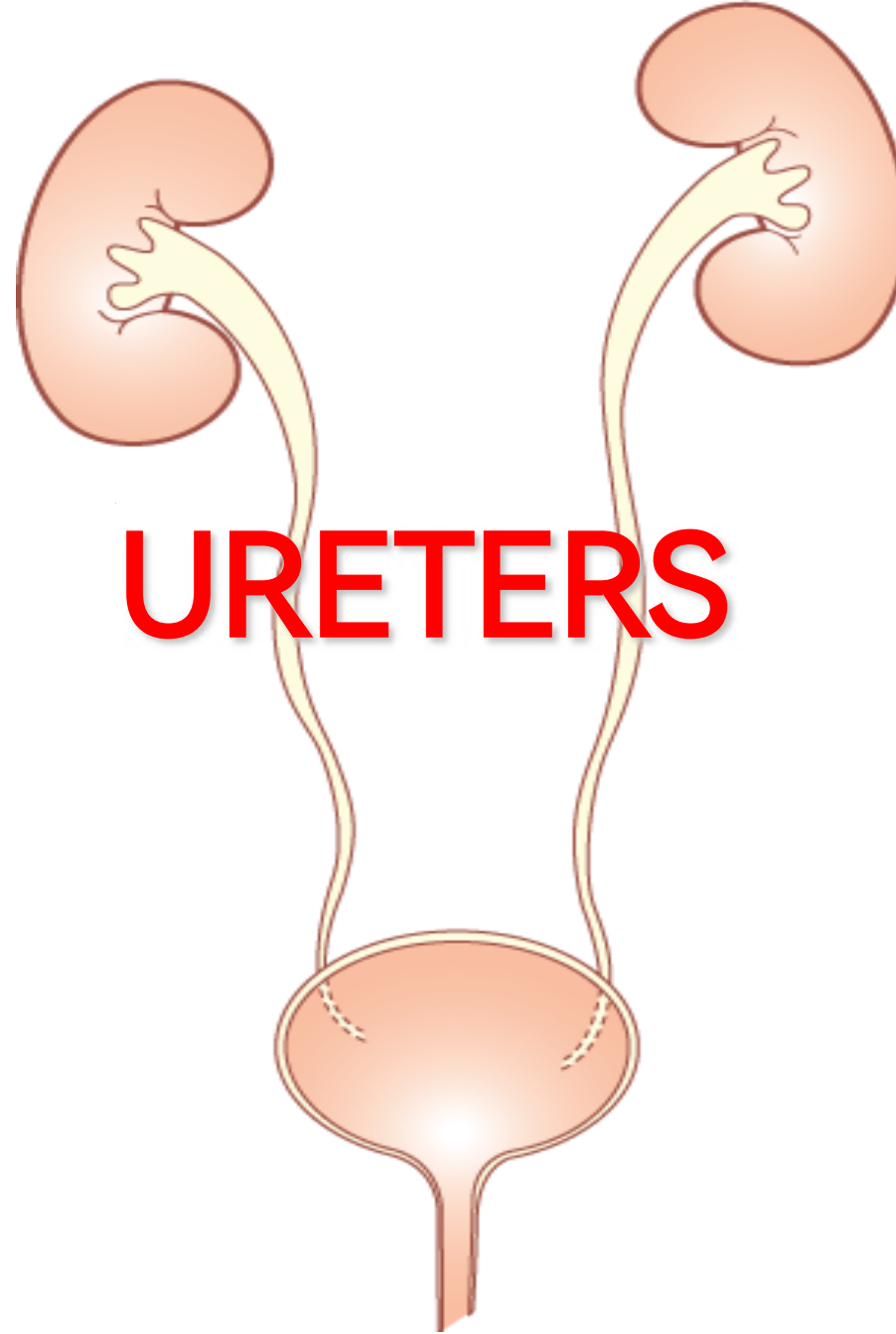


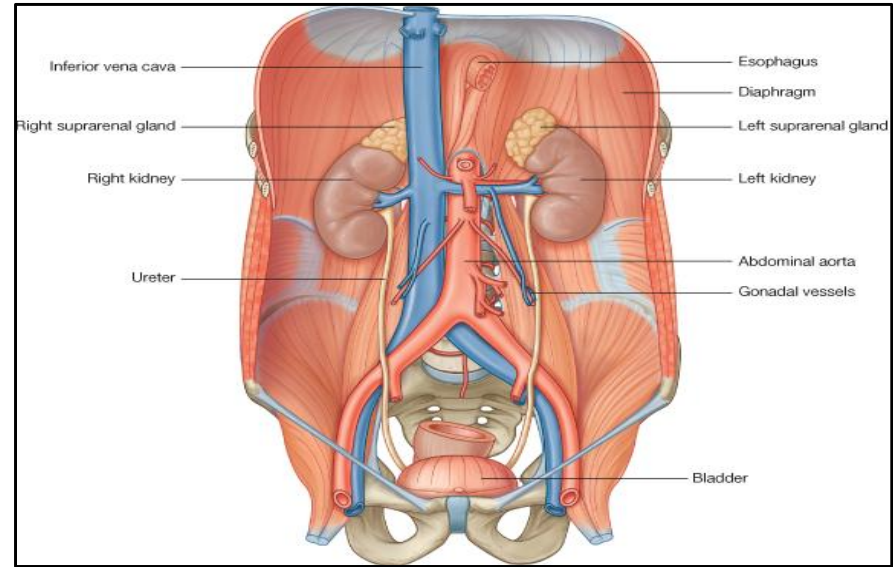


BY. M. G. HAMMAD



URETER

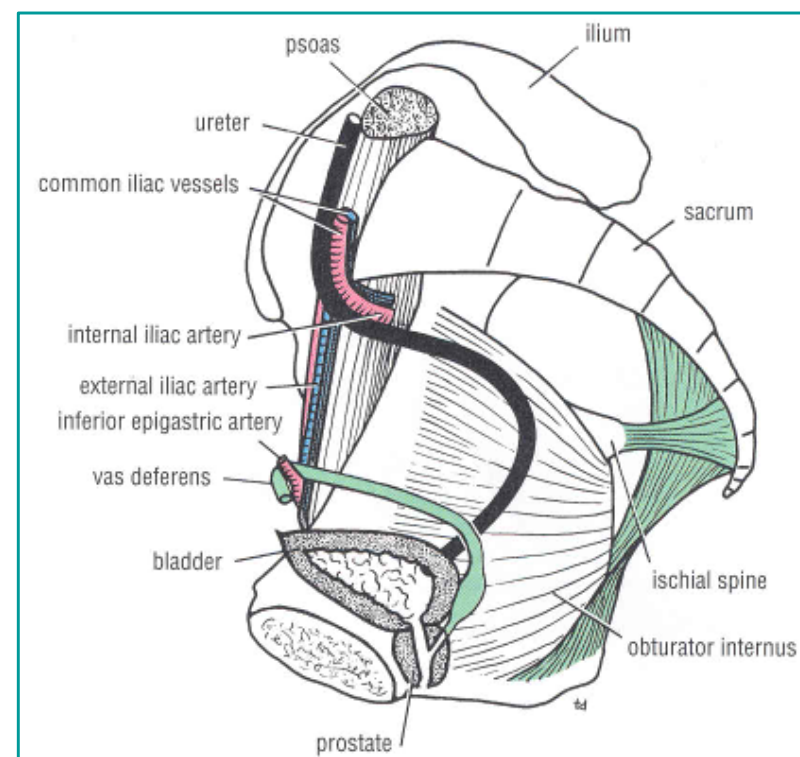
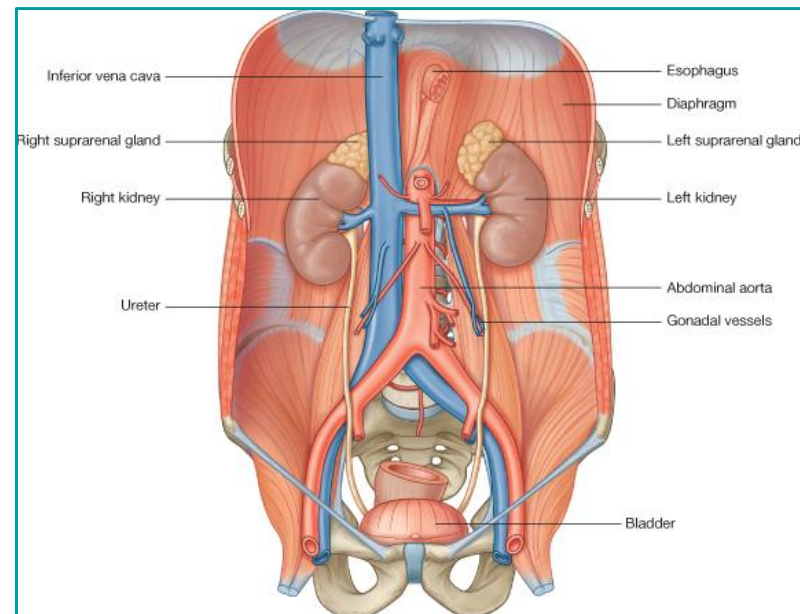
- A 25 – 30 cm long muscular tube.
- Transporting urine from kidney to urinary bladder.
- Begins as a continuation of renal pelvis.



- Course in abdomen:
 - It descends anterior to psoas major muscle (opposite the tips of lumbar transverse processes).
 - It crosses the end (bifurcation) of common iliac artery to enter the pelvis.
 - In abdomen it crosses:
 - ✓ The gonadal vessels (anterior to it).
 - ✓ The bifurcation of common iliac artery (behind to it).
 - ✓ Genitofemoral nerve (behind to it).

- Course in pelvis & termination:
- ❑ Runs downward in front to internal iliac artery, reaches ischial spine.
- ❑ Turns forward and medially, enters the upper lateral angle of urinary bladder
- ❑ Near its termination, is crossed by the vas deferens in male or uterine artery in female.
- ❑ Passes obliquely through the wall of bladder for about $\frac{3}{4}$ inch before opening into the bladder cavity.

❖ Bladder muscle contraction mechanically closes off ureteral orifice which prevents a reverse flow of urine toward the kidney.



RELATION OF URETER

➤ In Abdomen:

❑ Posteriorly:

- ✓ Psoas major muscle.
- ✓ Genitofemoral nerve.

❑ Anteriorly:

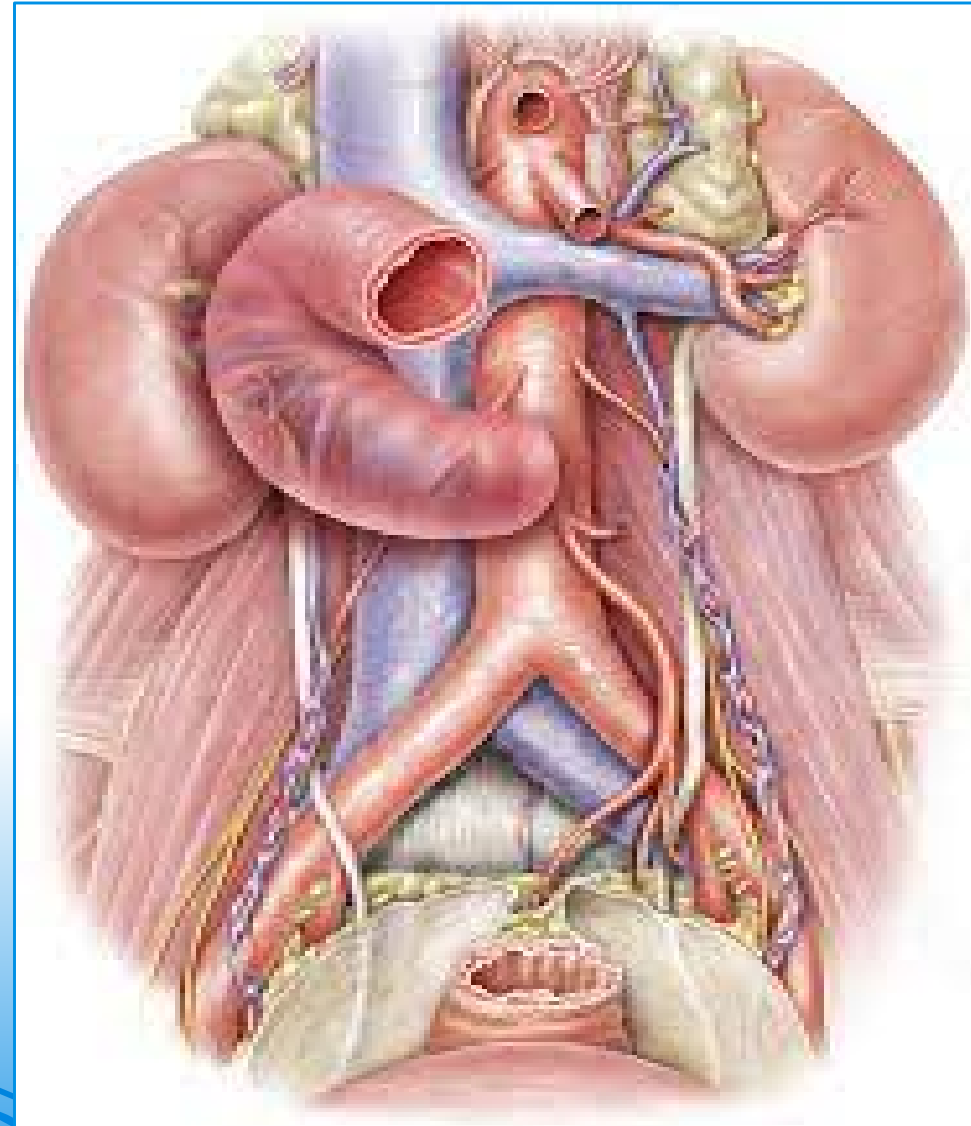
- ✓ Gonadal vessels.

❖ On right side:

- ✓ Duodenum.
- ✓ Root of mesentery.
- ✓ Right colic & ileocolic vessels.
- ✓ Superior mesenteric vessels.

❖ On left side:

- ✓ Left colic vessels.
- ✓ Apex of sigmoid mesocolon.
- ❖ IMV to the right of left ureter.



RELATION OF URETER

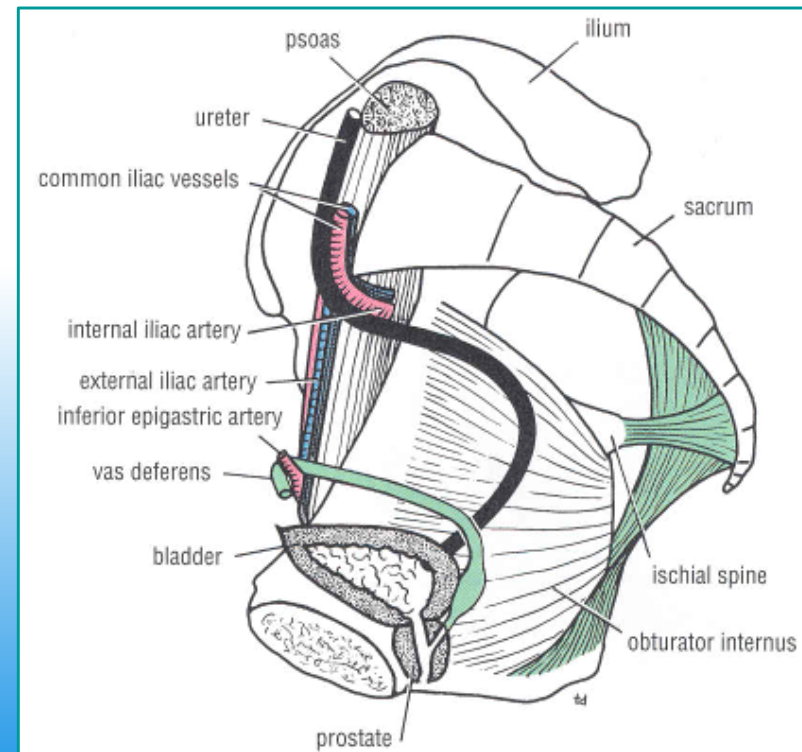
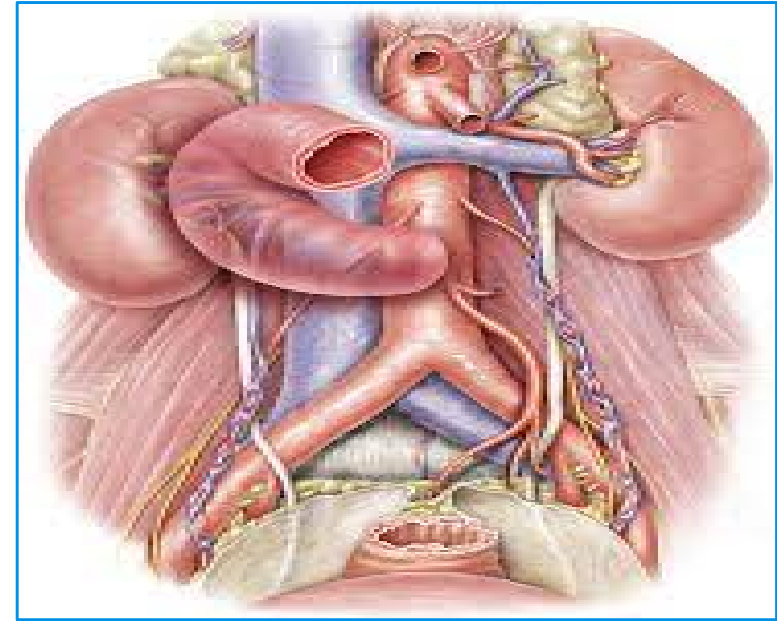
➤ In Pelvis:

❑ Posteriorly:

- ✓ Bifurcation of CIA.
- ✓ Genitofemoral nerve.
- ✓ External & internal iliac vessels.
- ✓ Obturator nerve.
- ✓ Superior vesical artery.
- ✓ Obturator vessels.

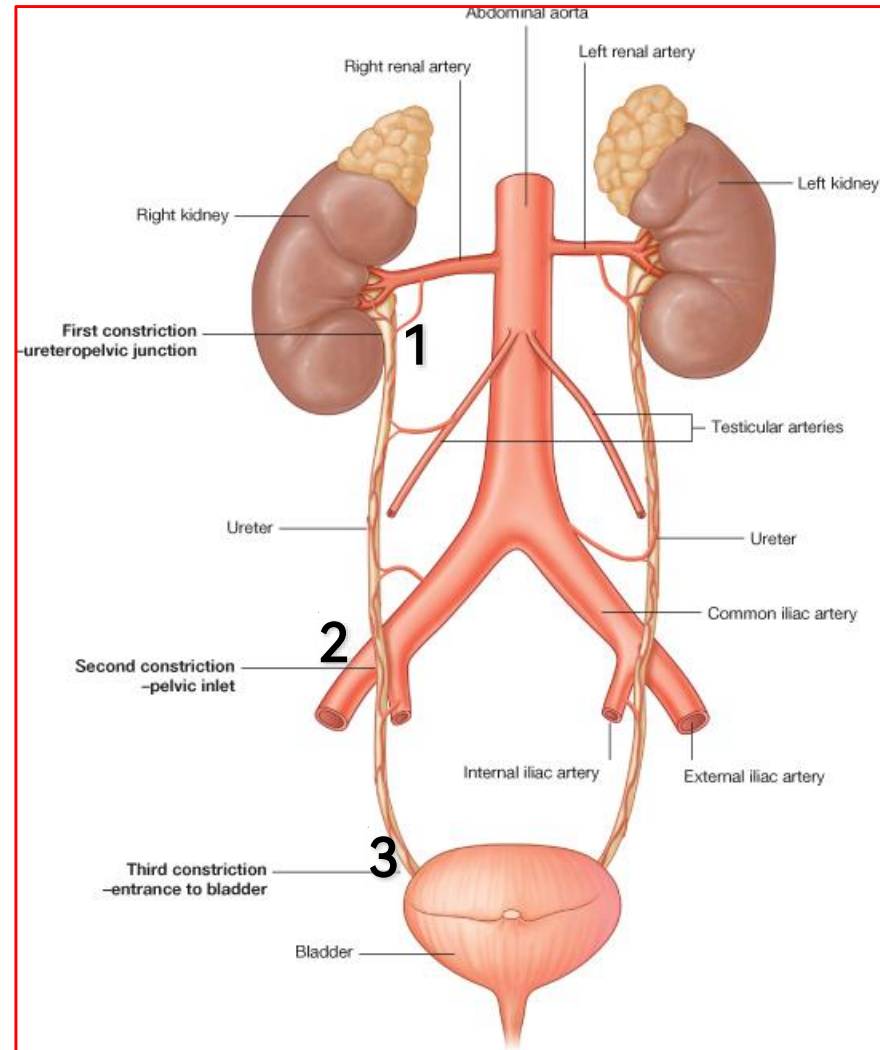
❑ Anteriorly:

- ✓ Ovaries (in female).
- ✓ Ductus deferens (in male).
- ❖ On the pelvic floor in the female, the ureter lies in the base of the broad ligament, here it is crossed superficially by the uterine artery.



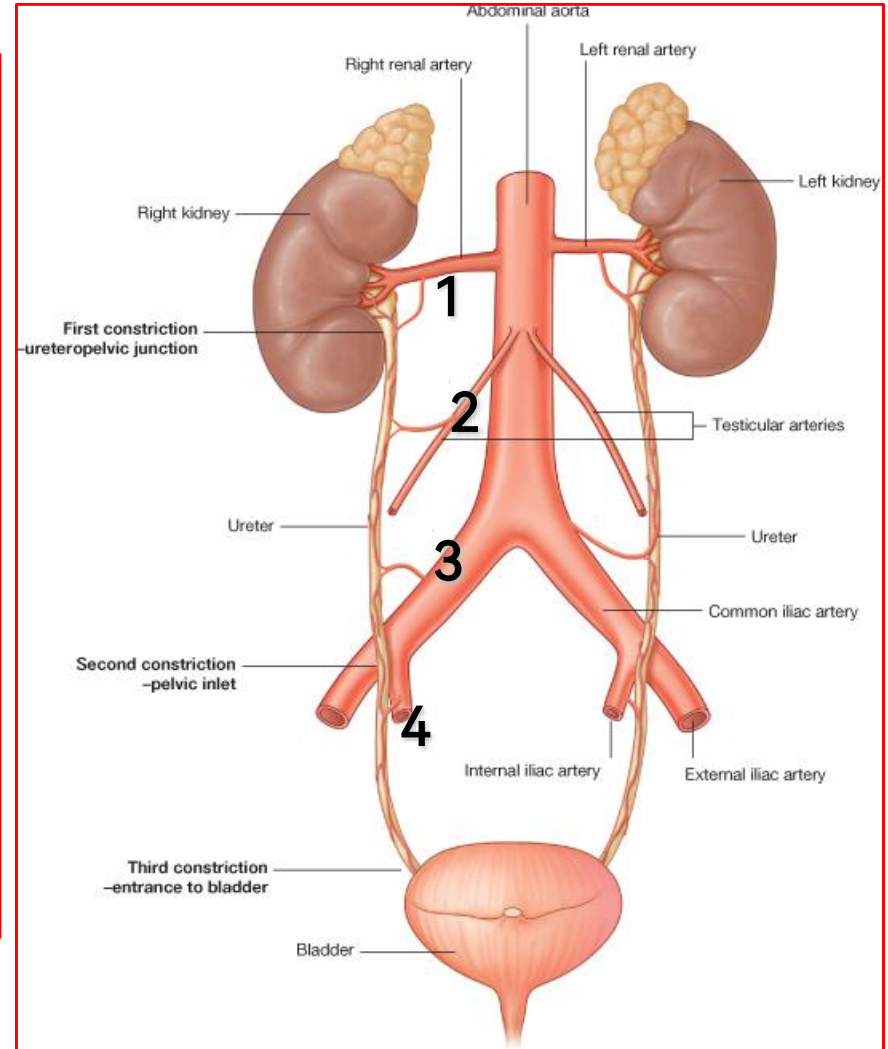
Ureteric Constrictions

- The ureter has constrictions at three points (sites of obstruction and stone impaction)
1. At the ureteropelvic junction.
 2. At the crossing of external/common iliac artery.
 3. At site of entrance to bladder.



Blood Supply

- Ureter is supplied by multiple arteries throughout its course
- From above downward, these are:
 1. Renal artery.
 2. Gonadal artery.
 3. Common iliac artery.
 4. Internal iliac artery.
- Veins corresponding to arteries.



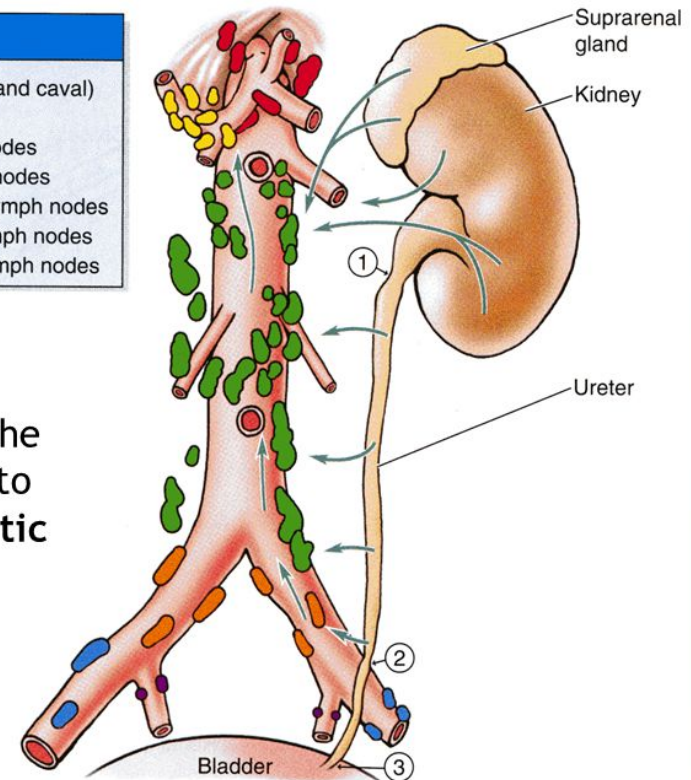
Lymph drainage

- Abdominal portion of the ureter drains into para-aortic nodes below the renal arteries.
- Pelvic portion into nodes on the side wall of the pelvis alongside the internal iliac arteries.

Key	
■	Lumbar (aortic and caval) lymph nodes
■	Celiac lymph nodes
■	Hepatic lymph nodes
■	Common iliac lymph nodes
■	Internal iliac lymph nodes
■	External iliac lymph nodes

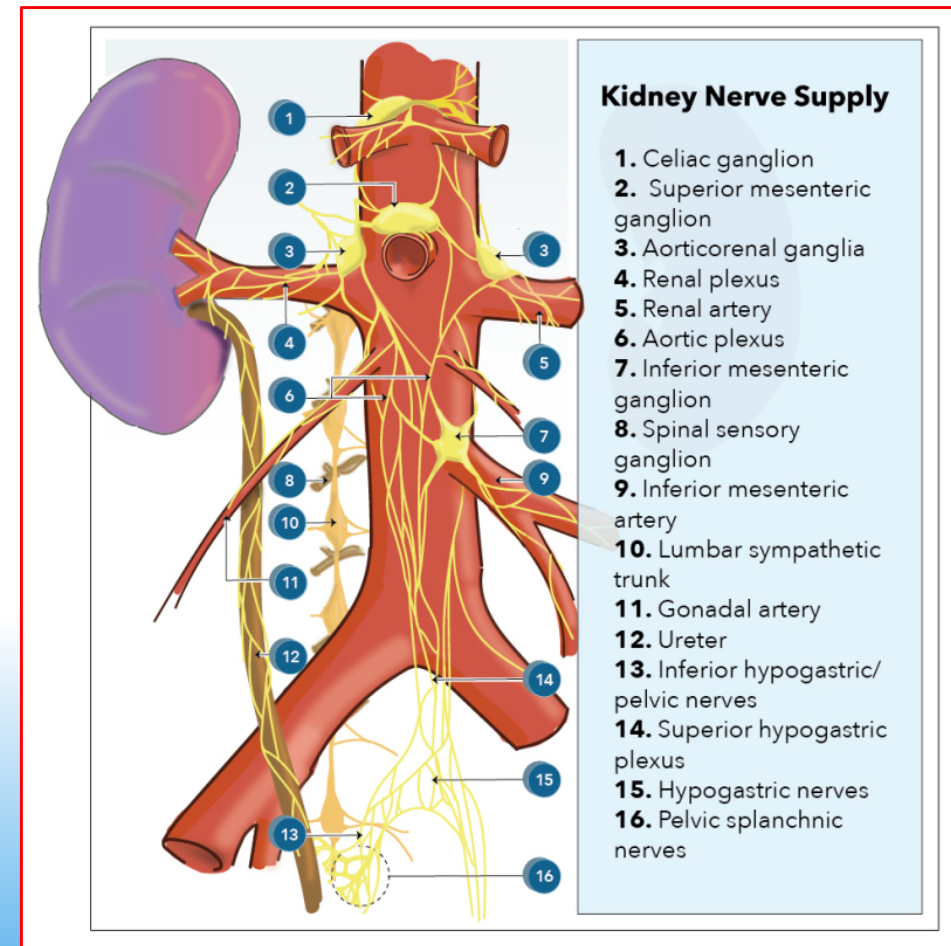
Ureter

The lymph of the ureters drains to the **lateral aortic nodes** and the **iliac nodes**.



Nerve supply

- Through autonomic NS:
- ❑ Sympathetic:
- ✓ From T1 1-L2 segments of via the coeliac and hypogastric plexuses.
- ❑ Parasympathetic:
- ✓ from the pelvic splanchnic nerves.
- ❖ Pain fibers accompany sympathetic nerves, as from the kidney



FLY
HIGH



Thanx & Good Luck

